EXP NO:14 EXPERIMENT TO UNDERSTAND FEATURE SCALING.

CODE :

```
import numpy as np
import pandas as pd
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler, MinMaxScaler
df = pd.read_csv('pre-process_datasample.csv')
df["Country"] = df["Country"].fillna(df["Country"].mode()[0])
features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values
age = SimpleImputer(strategy="mean", missing_values=np.nan)
salary = SimpleImputer(strategy="mean", missing_values=np.nan)
features[:, [1]] = age.fit_transform(features[:, [1]])
features[:, [2]] = salary.fit_transform(features[:, [2]])
oh = OneHotEncoder(sparse_output=False)
Country = oh.fit_transform(features[:, [0]])
final_set = np.concatenate((Country, features[:, [1, 2]]), axis=1)
sc = StandardScaler()
feat_standard_scaler = sc.fit_transform(final_set)
mms = MinMaxScaler(feature_range=(0, 1))
feat_minmax_scaler = mms.fit_transform(final_set)
print("Final Set:\n", final_set)
```

```
print("\nStandard Scaled Features:\n", feat_standard_scaler)
print("\nMinMax Scaled Features:\n", feat_minmax_scaler)
```

OUTPUT :

```
Final Set:
 [[0.0 0.0 0.0 ... 1.0 56.0 62884.06]
 [0.0 0.0 0.0 ... 0.0 53.0 89816.41]
 [0.0 1.0 0.0 ... 0.0 44.0 65339.32052173912]
 ...
 [0.0 0.0 0.0 ... 0.0 55.0 18000.0]
 [0.0 0.0 0.0 ... 0.0 40.0 65339.32052173912]
 [0.0 0.0 0.0 ... 0.0 58.0 65339.32052173912]]

Standard Scaled Features:
 [[-2.86794495e-01 -3.10630372e-01 -4.46139852e-01 ...
2.13437475e+00
   1.04820386e+00 -1.18109252e-01]
 [-2.86794495e-01 -3.10630372e-01 -4.46139852e-01 ... -
4.68521286e-01
   8.25741604e-01  1.17745987e+00]
 [-2.86794495e-01  3.21926022e+00 -4.46139852e-01 ... -
4.68521286e-01
   1.58354850e-01  2.10004088e-15]
```

...

[-2.86794495e-01 -3.10630372e-01 -4.46139852e-01 ... -4.68521286e-01

9.74049772e-01 -2.27723766e+00]

[-2.86794495e-01 -3.10630372e-01 -4.46139852e-01 ... -4.68521286e-01

-1.38261485e-01 2.10004088e-15]

[-2.86794495e-01 -3.10630372e-01 -4.46139852e-01 ... -4.68521286e-01

1.19651202e+00 2.10004088e-15]]

MinMax Scaled Features:

[[0. 0. 0. ... 1. 0.80851064 0.38383841]

[0. 0. 0. ... 0. 0.74468085 0.61415782]

[0. 1. 0. ... 0. 0.55319149 0.40483525]

...

[0. 0. 0. ... 0. 0.78723404 0.]

[0. 0. 0. ... 0. 0.46808511 0.40483525]

[0. 0. 0. ... 0. 0.85106383 0.40483525]]